

AI-Driven Multimodal Sentiment Analysis for Mental Health Monitoring on Social Media: A Resource-Constrained Comparative Study

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Abstract—Social media posts often combine text and images, but most sentiment-analysis systems for mental-health-related monitoring rely on text alone, potentially losing complementary emotional signal carried by attached images. This paper reports a controlled, same-dataset, same-metrics comparison of text-only, image-only, and multimodal (text+image) sentiment classification on MVSA-Single, a real, publicly documented Twitter image-text sentiment dataset (4,511 labeled pairs after standard cleaning; 3-class: positive/neutral/negative). We evaluate a traditional TF-IDF + Logistic Regression text baseline, a frozen-DistilBERT text model, a CNN trained from scratch, and a frozen-ResNet18 image model, together with two multimodal fusion strategies (feature-concatenation and late/decision-level fusion). The best multimodal model (feature-concatenation fusion) achieves 64.4% test accuracy and 56.3% macro-F1, outperforming the best unimodal model (frozen DistilBERT: 61.6% accuracy, 54.6% macro-F1) and the image-only model (52.7% accuracy, 42.8% macro-F1). A modality-conflict analysis shows the gain concentrates specifically on the 53.3% of test posts where text and image models disagree (fusion: 57.1% accuracy on this subset vs. 50.0% for text-only and 33.3% for image-only), with a small regression on posts where modalities already agree. We report this work entirely from real experiments run on constrained consumer CPU hardware (no GPU, 8GB RAM), documenting the resource constraints encountered and the engineering decisions made in response, as a contribution to resource-constrained reproducibility in this research area. Consistent with the non-clinical scope of this study, all outputs are the dataset's own sentiment labels, not diagnostic claims.

Index Terms—Multimodal sentiment analysis, mental health monitoring, social media analytics, natural language processing, deep learning, DistilBERT, ResNet18, feature fusion, MVSA, resource-constrained machine learning.

I. INTRODUCTION

Text-only or image-only sentiment analysis of social media posts each miss information that the other modality carries; the full background and motivation, derived from the originating project synopsis, is provided in the accompanying technical report (PROJECT_REQUIREMENTS_ANALYSIS.md). This project investigates whether, and under what conditions, combining text and image information improves

sentiment-based classification, framed explicitly as mental-health-related sentiment monitoring support — not clinical diagnosis (see Section VIII).

II. RELATED WORK

A full annotated review of 14 verified sources is provided in the accompanying literature review (LITERATURE_REVIEW.md). Key threads: text-based mental-health classification increasingly uses domain-adapted transformers such as MentalBERT [1], but a 2024 systematic review [2] documents widespread representativeness and evaluation-methodology issues across this literature. Image-emotion recognition benefits substantially from transfer learning over from-scratch training on modest datasets [3]. Multimodal fusion has a mature taxonomy [4], [5], but — per our research-gap analysis — lacks a controlled, adequately sized, same-metrics text-vs-image-vs-multimodal comparison specifically on real social-media data; the closest located prior work [6] evaluates its social-media condition on only 100 pairs.

III. RESEARCH GAP AND QUESTIONS

A full analysis is given in RESEARCH_GAP.md. The primary research question (from PROJECT_REQUIREMENTS_ANALYSIS.md) is: *can an AI-driven multimodal framework combining textual and visual information improve sentiment-based mental health monitoring on social media compared with unimodal approaches?*

IV. DATASET

MVSA-Single [7] consists of real Twitter image-text pairs with human-annotated 3-class sentiment labels (positive/neutral/negative). This project uses the published train/dev/test split from CLMLF [8] — 3,611/450/450 samples — for comparability with that and other work using the same fold, rather than an arbitrary self-computed split. Images and text were obtained via a HuggingFace

community mirror after the original OneDrive distribution proved unavailable (HTTP 404) at acquisition time; the full provenance chain, label-semantics verification (empirically confirmed as 0 = positive, 1 = neutral, 2 = negative by inspecting sample texts per label), and licensing caveats are documented in DATASET_SELECTION.md. The training-set class distribution is positive 2,147 (59.5%), negative 1,088 (30.1%), and neutral 376 (10.4%) — a real, measured imbalance that motivates macro-F1 as the primary evaluation metric. A duplicate-hash leakage audit found a small amount of cross-split duplication inherited from the published split itself (56/4,511 texts and 10/4,511 images, arising from Twitter retweets); this is disclosed in REPRODUCIBILITY.md as a limitation of using a third-party split and is assessed as unlikely to materially affect results at this scale.

V. METHODOLOGY

The full architecture rationale is given in PROJECT_ARCHITECTURE.md. All models share the same train/val/test split and evaluation code (src/evaluation/metrics.py).

A. Text Pipeline

Cleaning (URL/mention normalization, emoji-to-text demojizing, lowercasing), followed by either (A1) TF-IDF (1–2 grams, 10k features) with class-weighted Logistic Regression, or (A2) frozen DistilBERT (distilbert-base-uncased) mean-pooled embeddings with class-weighted Logistic Regression.

B. Image Pipeline

Resize and normalize, followed by either (B1) a small 4-layer CNN trained from scratch, or (B2) frozen ImageNet-pretrained ResNet18 penultimate-layer embeddings with class-weighted Logistic Regression.

C. Multimodal Fusion

Primary strategy (C): feature-level concatenation of the A2 text embedding (768-dim) and B2 image embedding (512-dim), fed into a 2-hidden-layer MLP with dropout, trained jointly with class-weighted cross-entropy loss and selecting the best checkpoint by validation macro-F1. Ablation: late (decision-level) fusion, in which independent A2/B2 classifiers' predicted probabilities are combined by simple averaging and by a validation-tuned weighted average.

D. Why Frozen Pretrained Encoders

This choice is justified directly by this project's measured hardware constraints (CPU-only, 8GB RAM) and by the literature (Section II) — transfer learning with a frozen or lightly adapted encoder is both the literature-supported choice for modest dataset sizes and the

only practically tractable choice on this hardware. REPRODUCIBILITY.md reports measured per-sample embedding-extraction timing (DistilBERT: 49.5 ms/sample; ResNet18: 536 ms/sample on first pass, dropping to a few ms/sample on a cache-warmed second pass).

VI. EXPERIMENTAL SETUP

A fixed seed (42) is used throughout (src/utils/seed.py). Evaluation metrics are accuracy, precision/recall/F1 (macro and weighted), and the confusion matrix; macro-F1 is prioritized given the class imbalance described in Section IV. All numbers reported in Section VII are read directly from results/*.json, generated by the scripts under src/text_model/, src/image_model/, and src/multimodal/.

VII. RESULTS

TABLE I
RESULTS: TEST-SET PERFORMANCE BY MODEL

Model	Acc.	Macro-F1	Wtd-F1
A1 — TF-IDF + LogReg	60.9%	52.0%	61.9%
A2 — Frozen DistilBERT + LogReg	61.6%	54.6%	63.5%
B1 — CNN from scratch	47.6%	42.9%	49.6%
B2 — Frozen ResNet18 + LogReg	52.7%	42.8%	54.3%
C — Concat Fusion	64.4%	56.3%	65.4%
C — Late Fusion (tuned)	64.0%	56.4%	65.6%

Full confusion matrices and a per-class breakdown are given in ERROR_ANALYSIS.md. Notably, B1 (trained from scratch, at a reduced 96×96 resolution and 3 epochs due to measured CPU-only training-time constraints — see Section V and REPRODUCIBILITY.md) scores close to B2 on macro-F1 (42.9% vs. 42.8%) despite a clearly lower accuracy (47.6% vs. 52.7%). This is an interesting nuance: B1's training instability (its validation macro-F1 dropped between epochs 1 and 2 before recovering by epoch 3 — a real, reported instability, not smoothed over) suggests its best-epoch macro-F1 may partly reflect noisy, under-converged training rather than a robust feature-learning advantage. This result is consistent with, not contradictory to, the literature-grounded expectation (Sections II, V) that frozen transfer-learned features are the more reliable choice under this project's real hardware constraints — B2 achieves comparable or better results with a fraction of the training instability and compute.

Model comparison answers the primary research question affirmatively, but with nuance rather than a blanket win: multimodal fusion outperforms every unimodal model on every aggregate metric, but the improvement is not uniform across classes or subsets — see Section VII-A.

A. Modality Conflict Analysis

Full detail is given in MODALITY_CONFLICT_ANALYSIS.md. Text and

image models disagree on 53.3% of test posts (240/450). On exactly this subset, fusion clearly outperforms both unimodal models (57.1% vs. 50.0% text-only, 33.3% image-only) — the clearest evidence that fusion helps specifically by resolving genuine cross-modal disagreement, not just by averaging noise. On the 46.7% of posts where modalities already agree, fusion shows a small regression versus text-only (72.9% vs. 74.8%). Per class, fusion helps substantially on the majority (positive) class but underperforms text-only specifically on the minority neutral class (42.6% vs. 48.9%) — attributed to neutral's severe data scarcity (10.4% of training data), leaving little signal for the fusion head to learn a reliable joint boundary.

B. Ablation: Fusion Strategy

Feature-concatenation (early/joint) fusion and validation-tuned late (decision-level) fusion perform almost identically (64.4%/56.3% vs. 64.0%/56.4% accuracy/macro-F1) — neither strategy dominates the other on this dataset/scale, and both clearly beat unimodal models. The late-fusion weight search independently converges on $w_{\text{text}} = 0.80$, corroborating text's stronger standalone performance (Section VII).

VIII. DISCUSSION, ETHICS, AND LIMITATIONS

A. Non-Clinical Scope

All model outputs are the dataset's own sentiment labels (positive/neutral/negative) and are never relabeled as clinical categories. ETHICS_AND_PRIVACY.md gives the full discussion of privacy, consent, licensing, bias, and misuse-risk considerations specific to this dataset and application framing.

B. Key Limitations

(1) MVSA-Single's labels are general sentiment, not mental-health-specific — a documented, unavoidable scoping limitation given that no open, legitimately licensed multimodal mental-health dataset was found (Section IV, DATASET_SELECTION.md). (2) The dataset is English-language, 2015–16-era Twitter content; results do not generalize to other languages, platforms, or eras without further validation. (3) A small (approximately 1–2%) cross-split duplication exists in the published split used. (4) CPU-only compute constraints limited the from-scratch CNN baseline's training budget and resolution — itself a finding relevant to resource-constrained reproducibility (Section V, REPRODUCIBILITY.md), not just a caveat. None of these limitations are minimized.

IX. CONCLUSION AND FUTURE WORK

Within a real, resource-constrained CPU-only environment, this project provides genuine, not assumed, evidence that multimodal fusion improves sentiment

classification on real social-media image-text pairs — concentrated specifically on posts where modalities disagree, with measurable trade-offs elsewhere. Future work, explicitly out of scope here per PROJECT_REQUIREMENTS_ANALYSIS.md, includes multilingual extension, cross-modal attention fusion at scale, and validation against a genuinely mental-health-labeled multimodal dataset if and when one becomes openly available.

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Dataset: Niu, Zhu, Pang, and El Saddik (2016) [7]; split and labels: Li, Xu, Zhu, and Zhao (2022, CLMLF) [8]. Pretrained models: DistilBERT and ResNet18, accessed via Hugging Face Transformers and torchvision, respectively.

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